STARTER MOTOR RELAY (Continued)

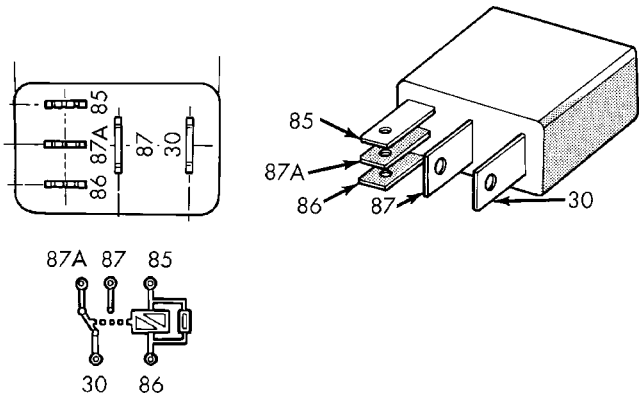

Fig. 14 Starter Relay

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

RELAY CIRCUIT TEST

(1) The relay common feed terminal cavity (30) is connected to battery voltage and should be hot at all times. If OK, go to Step 2. If not OK, repair open circuit to fused B(+) fuse in PDC as required.

(2) The relay normally closed terminal (87A) is connected to terminal 30 in de-energized position, but is not used for this application. Go to Step 3.

(3) The relay normally open terminal (87) is connected to the common feed terminal (30) in energized position. This terminal supplies battery voltage to starter solenoid field coil. There should be continuity between cavity for relay terminal 87 and starter solenoid terminal at all times. If OK, go to Step 4. If not OK, repair open engine starter motor relay output circuit to starter solenoid as required.

(4) The coil battery terminal (86) is connected to electromagnet in relay. It is energized when ignition switch is held in Start position. On vehicles with a manual transmission, the clutch pedal must be blocked in fully depressed position for this test. Check for battery voltage at cavity for relay terminal 86 with ignition switch in Start position, and no voltage when ignition switch is released to On position. If OK, go to Step 5. If not OK with a manual transmission, disconnect clutch pedal position switch wire harness connector and install a jumper wire between two cavities in body half of connector and check for battery voltage again at cavity for relay terminal 86. If now OK, replace faulty clutch pedal position switch. If still not OK with a manual transmission or if not OK with an automatic transmission, check for open or shorted fused ignition switch output (start) circuit to ignition switch and repair, as required. If fused ignition switch output (start) circuit is OK, refer to **Ignition Switch and Key Lock Cylinder**.

(5) The coil ground terminal (85) is connected to electromagnet in relay. On vehicles with a manual transmission, it is grounded at all times. On vehicles with an automatic transmission, it is grounded through park/neutral position switch only when gearshift selector lever is in Park or Neutral positions. Check for continuity to ground at cavity for relay terminal 85. If not OK with a manual transmission, repair open park/neutral position switch sense circuit to ground as required. If not OK with an automatic transmission, check for an open or shorted park/neutral position switch sense circuit to park/neutral position switch and repair, as required. If park/neutral position switch sense circuit checks OK, refer to **Park/Neutral Position Switch**.

REMOVAL

(1) Disconnect and isolate battery negative cable.

(2) Unlatch and open cover on Power Distribution Center (PDC) (Fig. 15).

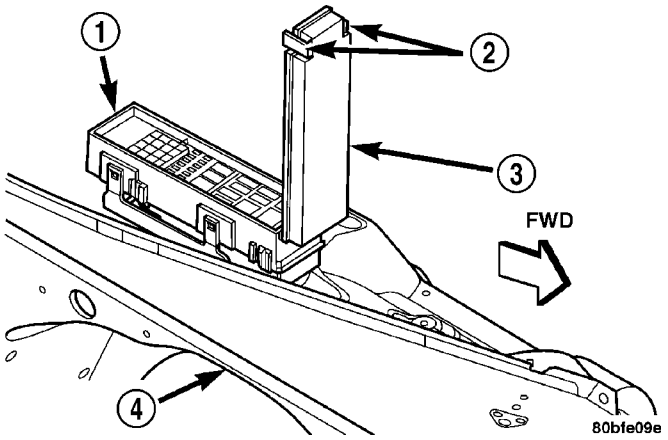

Fig. 15 Power Distribution Center

1 - POWER DISTRIBUTION CENTER (PDC)
2 - LATCHES
3 - COVER
4 - RIGHT FRONT FENDER

(3) See fuse and relay layout label affixed to underside of PDC cover for starter relay identification and location.

(4) Remove starter relay from PDC.

INSTALLATION

(1) See fuse and relay layout label affixed to underside of PDC cover for proper starter relay location.

(2) Position starter relay in proper receptacle in PDC.

(3) Align starter relay terminals with terminal cavities in PDC receptacle.

(4) Push down firmly on starter relay until terminals are fully seated in terminal cavities in PDC receptacle.

(5) Close and latch PDC cover.

(6) Connect negative battery cable.